

ECON 4538 - Problem Set 2 - Fall 2026

University of Minnesota

Due: October 5th 11:59pm

Instructions

Please answer the following questions using a Quarto/Jupyter notebook (either PDF or HTML format). For mathematical questions (LS questions), handwritten solution is accepted, provided it is clear to read and is properly merged into the notebook. For coding questions, please use Julia language. Submit the file in the respective Canvas assignment. Be clear and explain the steps you are taking, specially in the coding questions. **Also, include at the end a small section where you describe how you used AI tools to support you during this Problem Set.**

Question 1 (15 pts): LS Exercise 6.1

Question 2 (20 pts): LS Exercise 6.3

Question 3 (15 pts): LS Exercise 6.4

Question 4 (50 pts)

Part 1 (10 pts): Consider the McCall job search model with firing from LS 6.3.5. Let the wage space be uniformly discretized in a grid between 0 and 1 with 50 points on it. Program a function that receives as arguments (β, c, α, F) where

- β is the discount factor
- c is the unemployment compensation
- α is the exogenous firing probability
- F is a 50-size vector, where element i of this vector correspond to the probability of drawing the i -th element of the wage grid.¹

The function must return 3 elements estimated through value function iteration:

- The value function (50-size vector with the value function evaluated in each element of the grid)
- The policy function (50-size vector with 1 if *accept* or 0 if *reject*)
- The reservation wage

To show that the function works, plot the value function and the policy function for $(\beta, c, \alpha, F) = (0.95, 0.25, 0.1, (1/50, \dots, 1/50))$.

Part 2 (10 pts): Fix $\beta = 0.95$, $\alpha = 0.1$, $F = (1/50, \dots, 1/50)$. Plot the reservation wage against the unemployment compensation c for the interval $[0, 0.5]$.² Explain your results.

¹Thus its elements must be non-negative and must sum to 1

²For this, take a uniform grid with n (e.g. 50) values between 0 and 0.5, and apply the function that you created in the previous part using each of these values for c . You will have to save the values of the reservation wage for these, to then plot against the c grid.

Part 3 (15 pts): Fix $\beta = 0.95$, $c = 0.25$, $F = (1/50, \dots, 1/50)$. Plot the reservation wage against the exogenous probability of firing α for the interval $[0, 0.25]$. Explain your results.

Part 4 (15 pts): Fix $\beta = 0.95$, $c = 0.25$, $\alpha = 0.1$. Compare value functions for three distributions:

- The already used $(1/50, \dots, 1/50)$, where each element takes the value 0.2.
- Distribution vector where its first and last 10 elements take the value 0.03, and the 30 remaining values are $\frac{1-0.03*20}{30}$.
- Distribution vector where its first and last 10 elements take the value 0.04, and the 30 remaining values are $\frac{1-0.04*20}{30}$.

How do these reservation wages compare? Why?